



Materials Science 2

Electrical Properties of materials

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Aims

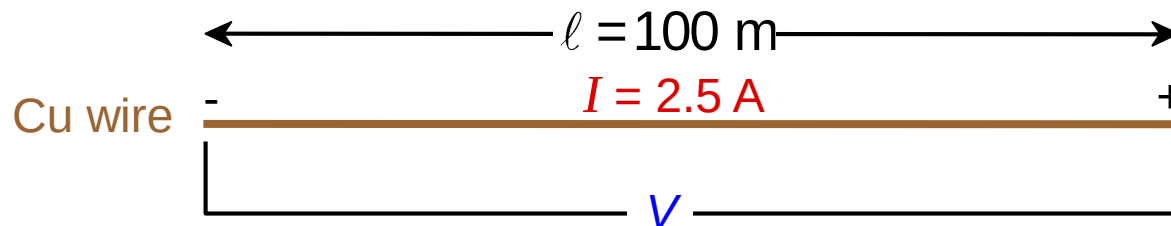
- Perform calculations of electrical properties of materials
- Calculate the electrical conductivities of metals
- Estimate electrical conductivity from mechanical properties
- Predict when a compound is an electron donor or acceptor
- Sketch variations in conductivity based on composition of alloys

Independent learning

- Use resources available (books, internet, lecture notes)
- Work with others: consult, brainstorm, share
- Read
- Try to solve the problem
- Ask questions

Worked Example: Calculating conductivity

What is the minimum diameter (D) that a Cu wire should be so that it conducts a voltage $V < 1.5 \text{ V}$?



$$R = \frac{\ell}{A\sigma} = \frac{V}{I}$$

Diagram illustrating the calculation of resistance R for a wire of length $\ell = 100 \text{ m}$. The resistance is given by $R = \frac{\ell}{A\sigma}$, where $A = \frac{\pi D^2}{4}$ is the cross-sectional area and $\sigma = 6.07 \times 10^7 (\text{Ohm}\cdot\text{m})^{-1}$ is the conductivity. The voltage V is less than 1.5 V and the current I is 2.5 A .

Solve to get $D > 1.87 \text{ mm}$

Example 1

Calculate the resistance of an aluminum wire 5 mm in diameter and 5 m in length. The conductivity of aluminum is $3.8 \times 10^7 (\Omega\text{-m})^{-1}$

- What would be the current flow if the potential drop across the ends of the wire is 0.04 V?
- What is the current density?
- What is the magnitude of the electric field across the ends of the wire?

Example 2

At room temperature the electrical conductivity and the electron mobility for aluminum are $3.8 \times 10^7 (\Omega \cdot \text{m})^{-1}$ and $0.0012 \text{ m}^2/\text{V} \cdot \text{s}$, respectively.

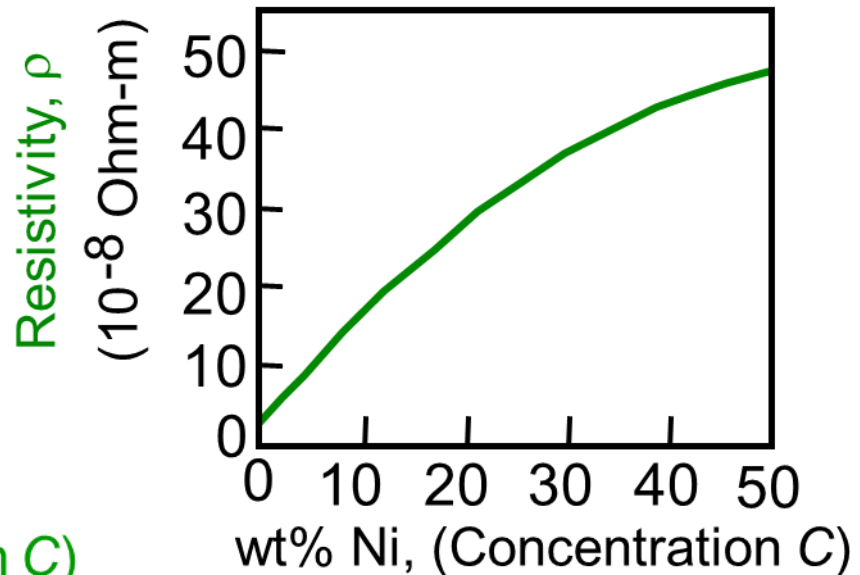
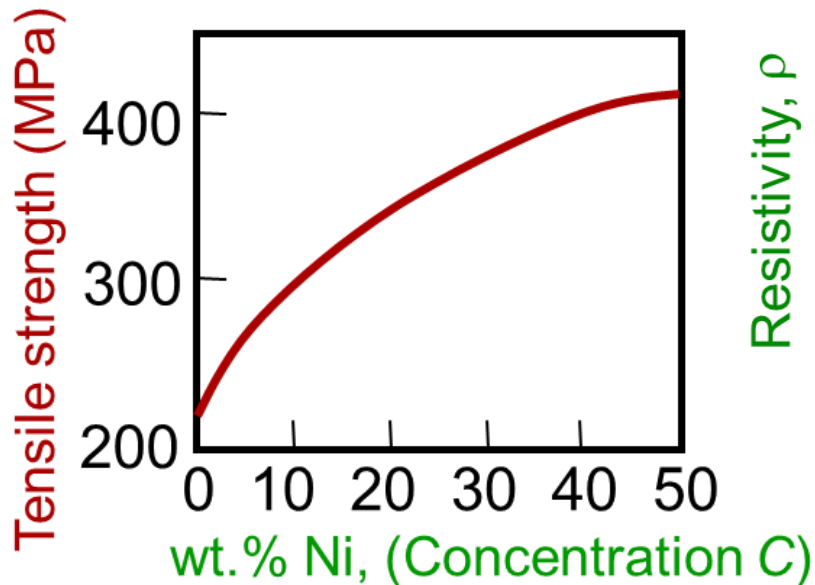
(a) Calculate the number of free electrons per cubic meter for aluminum at room temperature.

(b) What is the number of free electrons per aluminum atom? Assume a density of 2.7 g/cm^3 .

Example 3

Using the data provided in the figures below, determine the electrical conductivity of a Cu–Ni alloy that has a tensile strength of 275 MPa.

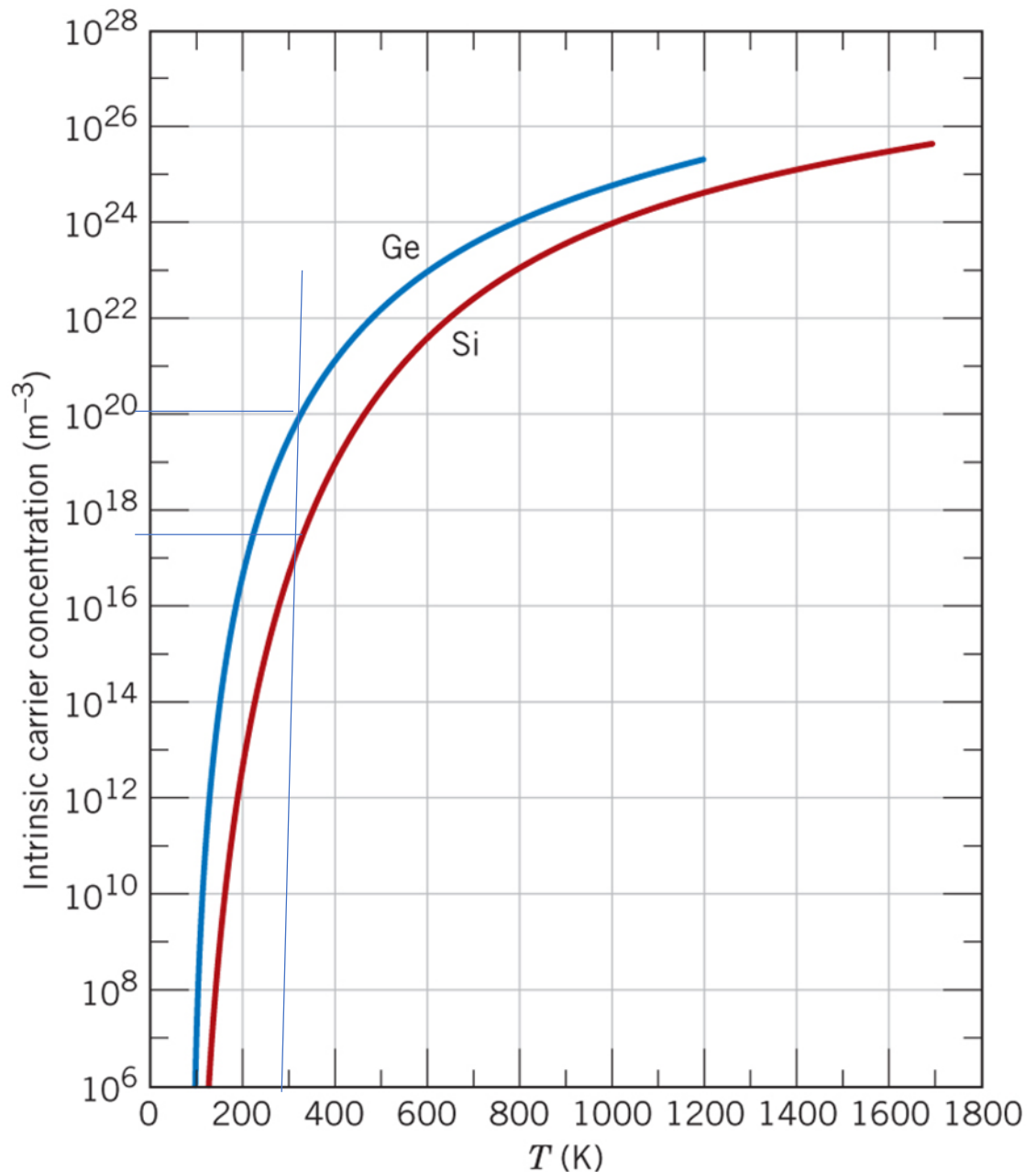
Adapted from Fig.
18.9, Callister &
Rethwisch 9e.



Example 4

(a) Using the data presented in the Figure, determine the number of free electrons per atom for intrinsic germanium and silicon at room temperature (298 K). The densities for Ge and Si are 5.32 and 2.33 g/cm³, respectively.

(b) Explain the difference in these values of free-electron-per-atom.



Example 5

At room temperature the electrical conductivity of PbS is $25 (\Omega \cdot \text{m})^{-1}$, whereas the electron and hole mobilities are 0.06 and $0.02 \text{ m}^2/\text{V} \cdot \text{s}$, respectively. Calculate the intrinsic carrier concentration for PbS at room temperature.

Example 6

For each of the following pairs of semiconductors, decide which has the smaller band gap energy, E_g , and then cite the reason for your choice.

(a) C (diamond) and Ge

(b) AlP and InAs

(c) GaAs and ZnSe

(d) ZnSe and CdTe

(e) CdS and NaCl

Example 7

Predict whether each of the following elements will act as a donor or an acceptor when added to the indicated semiconducting material. Assume that the impurity elements are substitutional.

Impurity	Semiconductor
N	Si
B	Ge
S	InSb
In	CdS
As	ZnTe

Example 8

At temperatures near room temperature, the temperature dependence of the conductivity for intrinsic germanium is found to be given by

$$\ln \sigma = \ln C - \frac{3}{2} \ln T - \frac{E_g}{2kT}$$

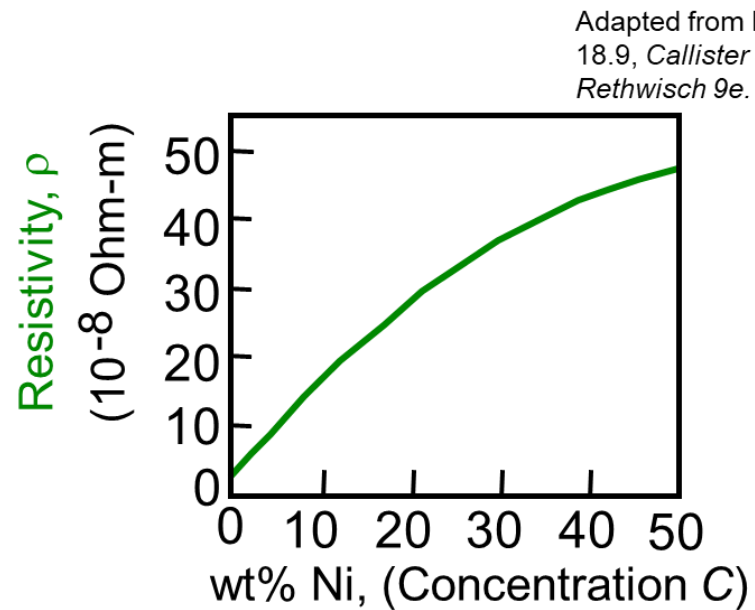
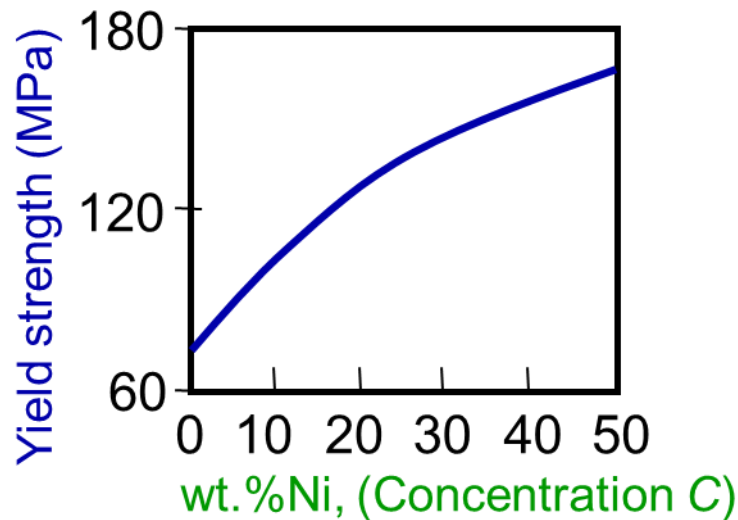
where C is a temperature-independent constant and T is in Kelvins. Using the above equation, calculate the intrinsic electrical conductivity of germanium at 175°C .

Example 9

Is it possible to alloy copper with nickel to achieve a minimum yield strength of 130 MPa and yet maintain an electrical conductivity of $4.0 \times 10^6 (\Omega \cdot \text{m})^{-1}$?

If not, why? If so, what concentration of nickel is required?

Use the data in the figures below.



Example 10

The room-temperature electrical resistivities of pure lead and pure tin are 2.06×10^{-7} and $1.11 \times 10^{-7} \Omega\text{-m}$, respectively.

- (a) Make a schematic graph of the room-temperature electrical resistivity versus composition for all compositions between pure lead and pure tin.
- (b) On this same graph schematically plot electrical resistivity versus composition at 150°C .
- (c) Explain the shapes of these two curves, as well as any differences between them.

Hint: You may want to consult the lead-tin phase diagram

If you finished all the examples...

Answer the following short text questions :

- Briefly describe electron and hole motions in a p–n junction for forward and reverse biases; then explain how these lead to rectification.
- State the differences in operation and application for junction transistors and MOSFETs.